

The Role of IT in Business Process-Oriented Organisations

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Abstract

The scope of this work is to investigate the role of information technology (IT) in business process-oriented (BPO) organisations. Moving away from the traditional organisational structures based on functional departments towards the adoption of a Business Process Orientation, organisations have to consider the significant effect on IT/IS systems designed to serve individual departmental requirements. The aim of this paper is to propose a model for the role of IT/IS supporting companies moving towards becoming BPO organisations. The model broadly encompasses three levels. The first level stresses the need for an appropriate strategy portfolio that will be formulated based on the principles of the business process concept. The second level highlights a portfolio of IT/IS features that Business Process-Oriented organisations should seek. Finally, the third level proposes some technologies and applications that can assist the realisation of the IT/IS features of the second level.

Keywords: Business Process Orientation, IS/IT modeling, Strategy Alignment

1. Introduction

Historically, the functional organisation evolved because it was the only model then known (i.e. from 19th century onwards) that was suitable for a large organisation [Walford, {1999}, Kirchmer, {1999}]. Contemporary business thinking questions this model of organisational structure, arguing that in the context of today's global and competitive market environment where customer satisfaction is the issue, such a structure cannot serve business requirements. It was extensively argued that organisations have to shift their focus from the strict functional hierarchies to horizontal business processes, which are the agents delivering value to the customer. In this sense, organisation-wide coordination is the major concern, rather than functional expertise. An in-depth glance reveals that business processes are multi-dimensional. Moreover, the concept of "business process-orientation" is also a multi-dimensional one. [Hammer and Champy (1993), Davenport (1993), Hammer (1996 and 1999), Armistead and Rowland (1996), Boehm and Phipps (1996), Love, Gunasekaran, and Li (1998), Day (1999), Kirchmer (1999), McCormack (1999), and

Ostroff (1999)] are some of the authors, researchers, and business consultants that dealt with the BPO organisation concept and discussed its business value.

Throughout the literature, it is very often highlighted that organisations shifting their focus from the traditional functional structure to a business process-oriented one, keep facing problems regarding legacy IT/IS. Existing IT/IS, as they are developed within the functional structure, appear to raise huge barriers to the adoption of a business process focus, considering certain features and weaknesses such as: 1. Lack of an alignment between corporate strategy and IT/IS development strategy, 2. Lack of integration and interoperability, 3. Department-based “islands of automation”, and “stand alone systems”, 4. Problematic and expensive maintainability, 5. Huge complexity, 6. Excessive, non value-added information processing times, 7. Outdated information, and 8. Loss of valuable information and knowledge. Such problems have been acknowledged in relevant published real case studies, [(Ross, 1999), Thoburn et al, {1999)].

However, it is also argued that in the near past (before 1990s), IT/IS couldn't offer the huge possibilities that can be offered today, and, in this sense, organisations that would choose to adopt a business process focus in the past (e.g. twenty years ago), would have struggled and, potentially, fail. This concept has to do with the trade offs of the so-called information “reach” and “richness” [Walters et al, (1999), Day, (1999), Bidault et al, {1999}]. “Reach” refers to the number of people able to share the same information at the same time, and “richness” refers to the amount of information that can be transmitted at one time, the extent that this information can be customised, and the extent within interaction (dialogue) is possible [Walters et al, (1999), Day, (1999)].

Bearing in mind the multifaceted nature of business processes, and the relevant requirements of BPO organisations, especially as far as information issues are concerned (e.g. BPO requires cross-functional information sharing, real-time information availability, availability of unfiltered information to the end workers/process members, feedback information directly from the customer, etc.). it can be said that current IT, can be viewed as an enabler for business processes [Davenport, (1993), Darnton et al, (1997), Ostroff, (1999), Day, (1999), Love et al, (1998), Rockart and Short, (1991), Seltsikas, (1999), Bidault et al, (1999)]. The lack of appropriate IT can hinder a BPO attempt quite significantly; in fact, it would not be an exaggeration to suggest that if IT is not employed in an effective manner, any attempt to embark on implementing business processes would be likely to fail [Darnton et al, (1997)].

Further from the enabling role of IT for realising business process orientation, it should be mentioned that IT could also have a driving role in the discussed area. For example, an organisation could shift its focus from the functional structure

towards business processes just because there are powerful IT applications available. However, IT should exist within a support framework for doing business; business should not be organised according to the occasional availability of technology, but according to planned strategies and pursued goals [Chan, (2000)].

The problems regarding IT/IS, as they are designed and developed to serve functional departments, necessitate a different approach to IT/IS from BPO organisations.

In the context of this paper, a model for the role of IT/IS that is meant to support companies moving towards becoming BPO organisations has been designed and developed, based on the principles posed by the business process concept.

2. *The proposed Model*

The model that the authors would like to propose, as a means of support for IT/IS in a business process-oriented organisation, broadly encompasses three levels:

- The need of a well-planned overall business strategy, adjoined to well-aligned IT/IS strategies;
- The necessary/suitable features of IT/IS in a BPO organisation, and
- Specific IT/IS applications suitable for a BPO organisation.

The above levels are interdependent, connected to one another by “determining”, and “enabling” links (figure 1). Moreover, the linked levels are subject to “external” influences/risks. The rationale on which the development of the model is based is explained:

- The initial idea is that an organisation wishing to adopt a business process orientation needs to formulate a business strategy that will guide it towards this route (a *BPO business strategy*). This business strategy will have to be established on the fundamental principles posed by the “business process” and the “business process orientation” concepts.
- Subsequently, well-aligned IS/IT strategies have to come into the picture. These IS/IT strategies will be built around and get directions from the overall business strategy (i.e. the “focus on business processes” strategy). In an iterative relation, these IS/IT strategies will act supportively to the realisation of the overall business strategy. More specifically:
 - ✓ The IS strategy will determine *what is required*, i.e. “what to do with information, systems, technology, and how to manage applications from a business point of view” [Ward et al, (1996)].

- ✓ The IT strategy will determine *how it can be delivered*, i.e. “how technology is to be applied in delivering information and systems and ultimately competitive advantage into the firm” [Ward et al, (1996)].
- The suitable IS/IT features (features that will serve the requirements imposed by the nature and dimensions of business processes) will then accrue, and will be captured within the determined IS/IT strategic framework. Such features will be the essential elements that will support the realisation of IT/IS strategies.
- Once the suitable IT/IS features are captured, the appropriate IS/IT applications can be determined. Iteratively again, the appropriate IS/IT applications are those that will deliver the suitable IS/IT features in the BPO organisation, as well as those that will enable and support business processes.
- Finally, well-operated and well-defined business processes are required in order to attain an overall successful business process-orientation.

The aforementioned relationships are depicted in figure 1.

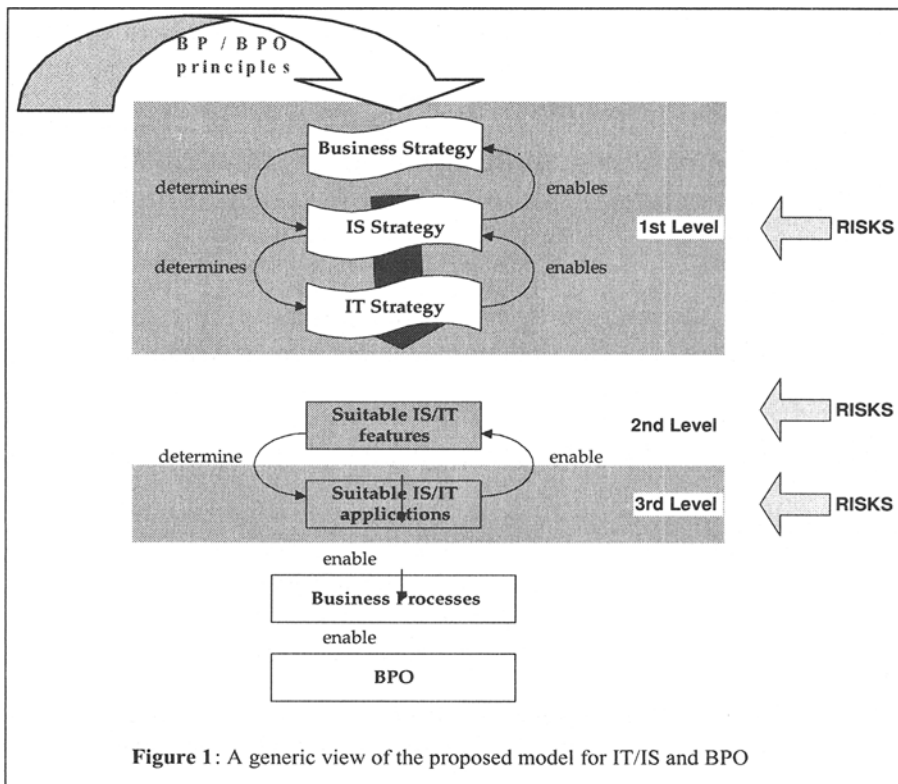


Figure 1: A generic view of the proposed model for IT/IS and BPO

2.1 Suitable IT/IS features in a BPO organisation

In an attempt to capture and determine the suitable features for IT/IS in the context of a BPO organisation, the authors have referred to the need for a strategic framework. Towards achieving a definitive approach to different features of IT/IS in a BPO organisation, business processes should be well understood and analysed. The key-requirements that cross-functional business processes impose, should encompass and address the following points:

- IT/IS should support the need for business process-wide **communication**, both internally (between and within process teams) and externally (encompassing customers and suppliers).
- IT/IS should support the need for **information sharing and exchange** along business processes (this is related to the previous point concerning communication). This is especially true when business process members are situated in different locations.
- IT/IS should support the **coordination** of different parts and members along business processes. Coordination is essential to ensure that all efforts are aligned towards the overall objective of the business process.
- IT/IS should support the need for **automation** of business processes. Davenport says that automation is the most commonly recognised benefit of information technology. IT has the ability to eliminate human labour and produce a more structured business process [Davenport, (1993)]
- IT/IS should support the organisation-wide **integration** between different applications. Integrated solutions can dramatically increase the speed of information delivery, streamline activities and decrease redundant data entry [Chan et al, (1999)].
- IT/IS should provide with **user-friendly interfaces**, since, in most cases, every individual that is a member of a particular business process will, more or less, have to interact with systems/applications on a regular basis.
- IT/IS should support the need for **real-time information** availability along business processes. Real time information is necessary to attain synchronisation, low cycle times and responsiveness in changing requirements.
- IT/IS should provide **performance feedback** directly to business process members. This will allow immediate improvements and adjustments of the

business process, according to customer satisfaction (this point relates to the previous one, referring to “real-time information”).

- IT/IS should support **performance measurement** in the organisation. This emerges from the fact that performance measurements are based on information about the deliveries of business processes. In this sense, it can be said that IT/IS is a business process **control mechanism**.
- IS/IT should be **flexible**, in terms of having the potential to adapt and change. This refers to the dynamic dimension of business processes. Business processes evolve and might change according to business requirements.
- IS/IT should be **centrally controlled, developed and maintained**. This is necessary, in order to address the complex integration issues and avoid unbalanced decentralised activities concerning IS/IT issues [Seltsikas, (1999), and Chan et al, (1999)]. This is a **key point**, highlighting that BPO moves away from the decentralised, departmental IT/IS systems. [Seltsikas (1999)], who investigated Xerox Ltd, a well-known example of a BPO organisation, has come to the conclusion that business process orientation implies complex integration, which may be managed with the use of modern IT, but it requires a particular kind of Information Management. In Xerox the Information Management elements are [Seltsikas, (1999)]: 1. Enterprise-wide IS, 2. Centralised business development, 3. Centralised on-going maintenance, 4. Centralised physical hardware (e.g. “server farms”), and 5. an Information Management group that is closely linked with the business that adopts business priorities.

IS/IT applications, which can satisfy the aforementioned IS/IT features for a BPO organisation, will now be highlighted.

2.2 Suitable IS/IT applications in a BPO organisation

The dramatic advances in IT/IS solutions and applications in the 1990s, especially the rapid emergence of the Internet has given a wide range of opportunities for BPO IS/IT support. The researchers have identified several solutions that can be deployed in the context of BPO, covering several of the required suitable IS/IT features. Such types of solutions will be presented in the following framework: 1. Technologies/applications that concern business processes within the organisation’s boundaries (*intraorganisational* systems), and 2. Technologies/applications that concern business processes extending beyond the organisation’s boundaries to involve customers and suppliers (*interorganisational* systems).

2.2.1 Intraorganisational systems

Within this category, the following technologies/applications have been identified to be relevant with the BPO concept and organisation: 1. Client/server applications, 2. Enterprise Resource Planning (ERP) Systems, 3. Data Warehousing, 4. Groupware, 5. Intranets, 6. Group Support Systems (GSSs), and 7. Workflow Management Systems (WFMSs)

2.2.2 Interorganisational systems

Within this category, the following applications of Information Technology were identified to be relevant with the “extended” BPO concept and organisation: 1. The Internet, and 2. Electronic Data Interchange (EDI)

3. Research Methodology

3.1 Design and structure of the questionnaire

The collection of some real world evidence was necessary in order to validate the proposed model and make it as rigorous as possible. The researchers decided to conduct a survey research, approaching the issue in a rather unique manner. The uniqueness lies on the fact that only one interview was conducted. The interviewee was an IT expert, that used to work for several major IT companies that implemented solutions for BPO organisations. The aim was to gather empirical evidence for several cases of BPO organisations and their approach to IT/IS issues. Data regarding seventeen cases of large BPO organisations, in which the interviewee had been involved in and could provide information, was collected.

A structured type of interview was followed. The questionnaire that was used to conduct the interview was an extensive one (16 pages), comprising thirty-six pairs of questions (a total of seventy-two questions). All of the questions were closed-ended in order to allow for relatively easy coding and process of the data, and easy group comparisons [Oppenheim, (1992)]. The first question of each pair simply asks for a quantitative answer, while the second one that asks for the opinion of the expert (qualitative data), is designed to have a scaled answer. The questionnaire was designed into three different sections. Each of these sections addresses one of the three different levels of the model, and aims at collecting data to test the corresponding level. Following, the three sections of the questionnaire are outlined.

The **first section** of the questionnaire comprises the set of questions one to eight, and the aim is to collect data that will measure the strategy approach of BPO

organisations towards IT issues, as it is presented in the model (figure 2). More particularly, our research effort is to find out whether BPO organisations formulate a well-defined, IT strategy integrated with an equally well-defined IS strategy, which, in turn, integrates with an overall business strategy that commences from the business process concept (questions one to six). Moreover, our aim is to get an idea whether organisations have perceived IT as a driver or as an enabler towards business process orientation (questions seven and eight). In this section, large BPO organisations and small/medium ones are examined separately because it is of interest to see potential differences in their strategy approach.

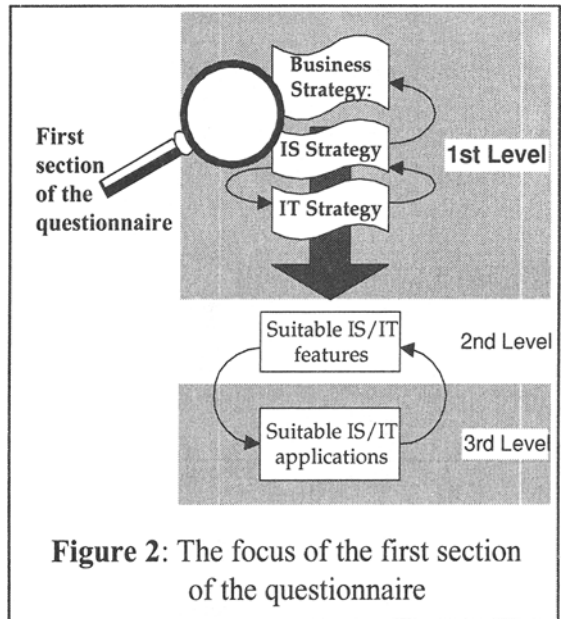


Figure 2: The focus of the first section of the questionnaire

The **second section** of the questionnaire refers to the second level of the proposed model (figure 3). It comprises the set of questions nine to twenty-two, and the aim is to investigate whether BPO organisations seek IT solutions that support issues like communication, information exchange, co-ordination, simplification, automation, real-time information etc. This section of the questionnaire does not distinguish large BPO organisations from small/medium ones, since we perceive that there is no reason for differentiation.

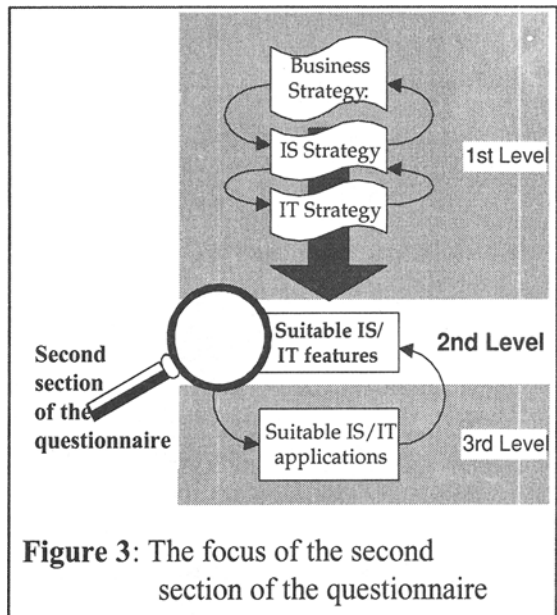
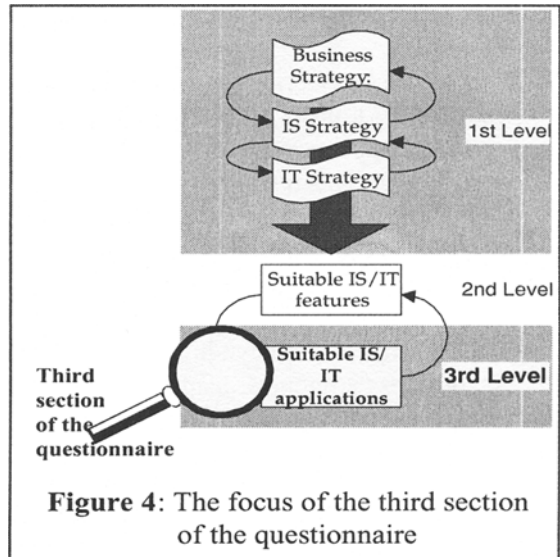


Figure 3: The focus of the second section of the questionnaire

The **third section** of the questionnaire refers to the third level of the proposed model (figure 4). It comprises the set of questions twenty-three to thirty-six, and the aim is to investigate whether BPO organisations use technologies and applications such as ERP applications, client server technology applications, Internet, Internet-enabled EDI, etc.



4. Research Results / Discussion

The aim and anticipation of this research effort was to collect real world evidence that will highlight how BPO organisations perceive and approach IT/IS issues and, also to test/validate the proposed model. The analytical presentation of the empirical evidence that will follow is structured around each of the three levels of the proposed model.

4.1 First level of the model (BPO, IS/IT & Strategy)

On the empirical level, and referring to the seventeen cases discussed during the interview, figure 5 summarises the collected data

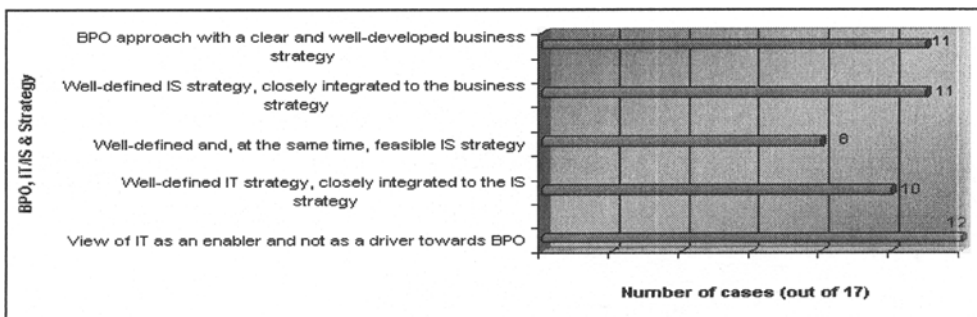


Figure 5: Empirical data referring to the first level of the proposed model

The points of interest at this level are presented and discussed below:

1. Eleven large organisations, while in the process of adopting a BPO, appeared to have a clear and well-developed business strategy. Most of these eleven companies were based in Western Europe, and some of them in Eastern Europe. The remaining six, were approaching the transformation to BPO, lacking an appropriate strategic business framework. These six companies, as it was revealed from a follow-up question, were all based in Eastern Europe. This fact is related to the volatile political, business and economic environment in Eastern Europe during the early and middle 1990s. This volatility might have hindered the formulation of appropriate, long-term BPO business strategies, leading Eastern Europe organisations to adopt quick-fix, short-term ones that would help them survive.
2. The interviewee stated that the same eleven organisations, have had a well-defined IS strategy that was closely integrated to their BPO business strategy. In the same fashion as above, the remaining six that lacked such a strategic framework for IS were Eastern Europe ones. Additionally, the interviewee's perception was that among the eleven BPO organisations that had a well-defined IS strategy, only in eight cases this IS strategy was a feasible one. As the interviewee had stated, part of his role was to influence the development of a well-defined and well-aligned IS strategy in these organisations. A well-defined IT strategy was absent in the six Eastern European BPO organisations that had neither a well-developed BPO business strategy, nor a corresponding well-aligned IS strategy. Additionally, from the eleven BPO organisations that were satisfactory as far as their business and IS strategies were concerned, ten appeared to also have a well-defined IT strategy, closely integrated to their IS strategy. One company, based in Eastern Europe, had problems with this issue: while proper business and IS strategies were present and well-aligned, the IT strategy was problematic.
3. According to the interviewee, twelve large organisations viewed IT as an enabling agent towards BPO. The remaining five, perceived IT as the driver that would lead them to BPO. A comment at this point could be that those organisations that, generally have had a well-defined strategic approach, perceive IT as an enabler towards BPO. On the other hand, Eastern Europe organisations that were facing big problems during most of the 1990s, and have approached BPO as a survival type of strategy, viewed IT as the agent that would drive through change.

The collected data, regarding the particular level of the model (*BPO-IS/IT-Strategy*), indicates that BPO organisations appreciate the need for approaching IS/IT issues through a well-planned, BPO strategy framework, and they successfully perceive IT

to be an enabler and not a driver towards BPO. Even though it appears that in six out of the seventeen cases such a strategic framework was not present, and IT was treated as a driver towards BPO, the previous comment can be perceived to be a valid one. This can be true, considering Eastern Europe's problematic/volatile environment during most of the 1990s, which probably drove these six organisations towards quick-fix, short-term strategies.

A further comment is that, although under normal circumstances an overall strategy portfolio seems to be commonplace, there is an indication that the formulation of feasible IS strategies sometimes is problematic. The researchers believe that this could refer to possible discrepancies between the IS needs of the BPO organisation and the IT applications available at the time. It might be that the IS strategy is justified and well-developed, but if the appropriate IT applications are not available, or they are extremely costly, the IS strategy might be seen as unfeasible. It is likely that unfeasible BPO IS strategies refer to cases in the early 1990s, when IT applications that could support BPO were either emerging, or very expensive.

Finally, it should be mentioned that the interviewee completely agreed that a strategy framework on behalf of BPO organisations is necessary when approaching IS/IT issues. Moreover, they highlighted that for Eastern European organisations that were facing big problems during the early and middle 1990s, IT was acceptably viewed as a way to drive through change and approach BPO.

4.2 Second level of the model (Suitable IS/IT features)

The second level of the proposed model refers to the eleven features that BPO organisations should seek. IS/IT in a BPO environment should be able to support communication, information exchange, co-ordination, real-time information, integration of existing systems/applications, performance measurement, and simplification and automation of business processes. Moreover, IS/IT should be flexible, have a user-friendly interface, and should be centrally controlled, developed and maintained. The relevant empirical data, referring to the seventeen cases discussed in the interview, is summarised in figure 6.

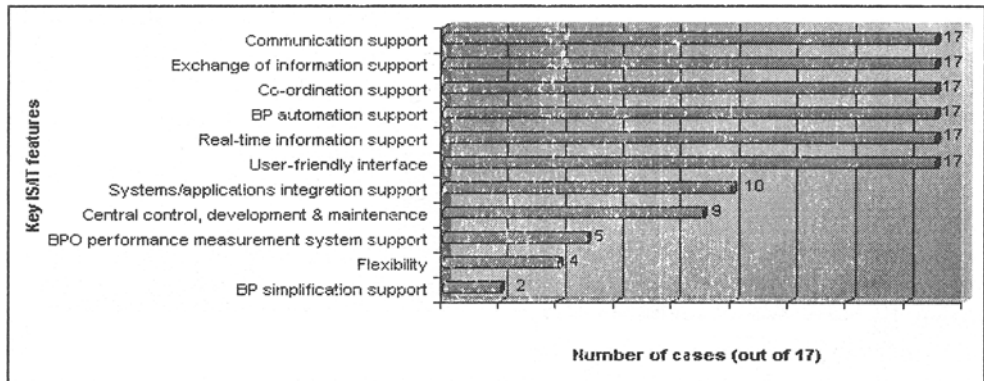


Figure 6: Empirical data referring to the second level of the proposed model

More specifically, the points of interest at this level are the following :

1. In all the available cases of large BPO organisations, six out of the eleven IS/IT features of the model were commonly sought: 1. Communication support through whole business processes; 2. Exchange of information support between different members and areas of a business process, 3. Co-ordination support between different members and areas of a business process, 4. Business process automation support; 5. Provision of real-time information for different members and areas of a business process, and 6. User-friendly interfaces.

A first comment here can be that the ideas of communication, exchange of information, and co-ordination were not explicitly perceived as part of IT/IS support for BPO. Most of the sample BPO organisations considered them a natural outcome of IT/IS deployment. However, the researchers believe that these three features should be explicitly sought, since there are no guarantees that the realised IS/IT would efficiently facilitate communication, co-ordination, and exchange of information within the business processes. Furthermore, automation of business processes was perceived as one of the major issues that IT/IS should facilitate and support..

2. Ten out of the seventeen sample BPO organisations sought IT/IS solutions that would support the integration of existing systems through and between business processes. The remainder seven BPO organisations did not conceive this as necessary. According to the interviewee, the latter was especially true in Eastern Europe, where BPO organisations wanted to deploy totally new systems. The researchers believe that this can be linked back to the previously discussed strategy issues. It seems that organisations that had a well-defined, appropriate

strategic approach to BPO and IT/IS, considered the possibility to hold on to their legacy systems, if these systems could be integrated and serve BPO needs. On the other hand, the sample BPO organisations that lacked an appropriate strategy portfolio, overlooked this possibility. The fact that they sought totally new systems highlights the previously mentioned suggestion that they perceived IT as the agent that would lead them to BPO and, as the interviewee said, solve all of their problems.

3. Nine BPO organisations of the sample, sought IT/IS solutions that could be centrally controlled, developed and maintained. The interviewee stated that, in each case, this was directly related to the occasional facilitator that drove the IT program. In the cases that the IT program was driven by process owners (formerly managers), IT/IS solutions that would be locally controlled, developed and maintained were chosen (this is true to eight cases of the sample). On the other hand, in the remainder nine cases, the IT program was facilitated by senior management. The interviewee stated that IT/IS systems should be centrally controlled, developed and maintained, to avoid complexities and make the best use of resources.
4. IT/IS as a support to a BPO performance measurement system, appears to be something that was not thought of, to a great extent. Moreover, it seems that an appropriate performance measurement system that would measure business process elements (e.g. cycle time, team performance, customer satisfaction etc), was not commonly present in the first place. Only five BPO organisations of the sample had formulated a proper BPO performance measurement system, and in these cases, IT was deployed to measure performance. This indicates that, to some extent, BPO organisations underestimate, or disregard what the researchers perceive to be a major issue for BPO: a BPO performance measurement system. This deficiency is probably related to a gap in the formulated business strategy, or even more, to an overall limited understanding of the business process concept.
5. Finally, only four BPO organisations of the sample sought IT/IS solutions that were flexible, in terms of being able to change and adapt when changes in business processes occur, and different requirements must be met. A reason for this, as the interviewee stated, might be that most BPO organisations tend to concentrate on achieving the current business process design, since they perceive it as a big and complex enough task. We believe that trying to add flexibility to IT/IS, can add to the complexity of the systems; nevertheless, they believe that within a long-term BPO strategy where changes in business process requirements are appreciated, IT/IS flexibility is a prerequisite.

4.3 Third level of the model (*Suitable IS/IT applications*)

The third level of the proposed model refers to what has been defined as “suitable IS/IT applications” for BPO organisations. Eight IT/IS applications have been proposed that large BPO organisations can deploy to support their business processes: general client/server technology applications, ERP, EDI, the Internet, Data Warehousing, Groupware, Workflow Management Systems, and Group Support Systems. This list is by no means an exhaustive one, since new applications and specialised commercial solutions are introduced with high frequency. However, we believe that the list encompasses the major applications and technologies that could support BPO to date. The relevant empirical data, referring to the seventeen cases, is summarised in figure 7.

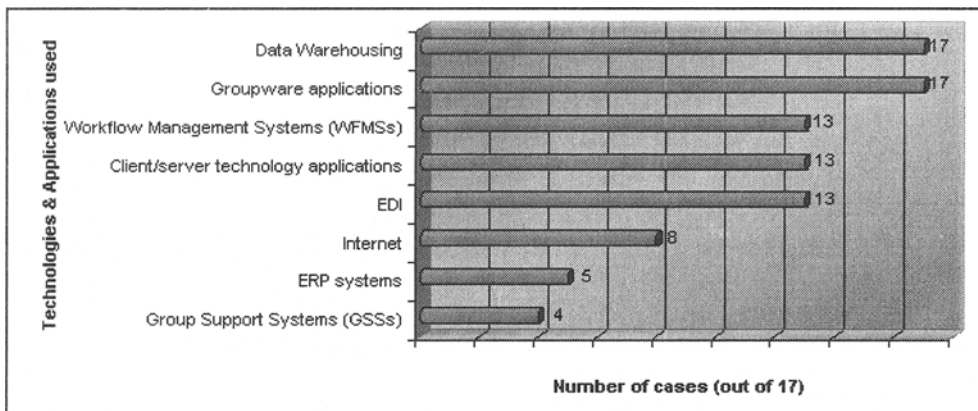


Figure 7: Empirical data referring to the third level of the proposed model

More specifically, the points of interest at this level are presented and discussed below:

1. In all the seventeen cases of the sample, Data Warehousing and Groupware applications were deployed. This seems consistent with the data in the first level, where in all of the seventeen cases, communication, information exchange, and co-ordination were IT/IS features commonly sought.
2. In thirteen cases of the sample, general client/server technology applications, Workflow Management Systems (WFMSs), and EDI were deployed. Although in all seventeen cases, the automation of business processes was perceived as important, in four of them WFMSs were not deployed. Either there is an inconsistency at this point, and sometimes BPO organisations fail to satisfy their already perceived individual requirements, or possibly, some

other systems were deployed towards automation, not identified in this work. In the four cases that EDI was not used to support communication with suppliers and/or buyers, there is an indication that these BPO organisations possibly did not perceive business processes to extent beyond the organisational boundaries to include suppliers/buyers. This comment can be valid, assuming that these BPO organisations could afford the financial cost of deploying such a system, which is substantially expensive.

3. The same can be highlighted as far as the use of ERP systems is concerned. Only five BPO organisations of the sample used ERP systems, as they are known today. An interesting comment here can be that in all these cases, standard ERP packages were chosen, and all five organisations modified their business processes to fit the ERP package. This is clearly an indication that the advantages of using an integrated, standard ERP solution are appreciated from BPO organisations. The current use of standard ERP systems is even more extensive, due to the availability of increasingly configurable standard commercial packages.
4. Although an answer was given (i.e. four BPO organisations used GSSs), comments on the use of Group Support Systems cannot be made, because the interviewee seemed uncertain and ambiguous about the issue. The same uncertainty and ambiguity was identified when the interviewee was asked about the use of Intranets in the cases he experienced.

5. Conclusions

This paper particularly focused on the role of IT/IS in the BPO environment. The shift away from the traditional functional structures, and towards BPO, entails serious implications on IT/IS. This is mainly because IT/IS, traditionally, have been deployed to serve the fragmented goals and information needs of individual functions/departments, while the principles of BPO imply a horizontal, cross-functional view in every aspect of the organisation.

Organisations, on their way to adopt BPO, need to consider IT/IS as an important factor, though on a different basis, compared with the traditional decentralised approach. It was argued that organisations must approach BPO and IT/IS issues through a strategy portfolio that is based on the fundamental principles of the BPO concept (BPO Business/IS/IT strategy). It was also argued that, based on this proper BPO strategy portfolio, BPO organisations should seek solutions with features that enable the organisation-wide coordination of people and technologies (e.g. integration support, real-time information support, etc). Finally, a set of applications/systems that could deliver these features was proposed as well.

The above elements were logically argued and linked to each other, on a theoretical level. Additionally, the exploratory empirical research, provided with indications that the proposed model concerning IT/IS and BPO organisations has certain value. It pointed out that, to a great extent, large organisations that have adopted BPO, seek IT/IS solutions that have the capabilities to satisfy what the researchers have addressed as “suitable IT/IS features for BPO”. Nevertheless, three issues that have been identified as extremely important, appear to be overlooked by BPO organisations: 1. Flexibility of systems/applications, 2. BPO Performance Measurement support, and 3. Central control, development, and maintenance of systems/applications (although indications for the latter are not as strong as for the first two). The latter is quite interesting; it highlights that IT in BPO organisations should go back to the main principle of the mainframe era (centralised IT).

An additional indicative conclusion is that BPO organisations appreciate the need for approaching IS/IT issues through a well-planned, BPO strategy framework, and they successfully perceive IT to be an enabler and not a driver towards BPO. However, there is an indication that there is a problem with the formulation of feasible IS strategies, that would provide with a structured approach towards the deployment of IT systems/applications.

Far from the above, it is our view that the future of BPO will be directly linked with a particular kind of IT. The evolution of the Internet is characterised with ever-increasing technical capabilities, in terms of security, reliability, and speed. It is no surprise that systems like ERP and EDI are currently developed using architectures that allow direct contact and operation over the Internet.

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